ULTRA HIGH-DEFINITION DIGITAL SURFACE MODEL AND GEOCHEMICAL MAPPING OF THE STEFANOS CRATER AT THE ACTIVE VOLCANO OF NISYROS IN GREECE WITH DRONE SCANNING, PORTABLE X-RAY FLUORESCENCE AND GIS

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Abstract

The Hellenic Active Volcanic Arc (HAVA) has resulted from the interaction of the rolling-back and subducting African plate and the overriding and extending Aegean microplate. At the western end of the HAVA, Nisyros volcanic island hosts the Stefanos crater with a history of numerous volcanic eruptions and associated faulting. Drone mapping at an altitude of 30 meters above the crater's surface has generated a digital surface model (DSM) and cloud densification of approximately 40 million points with an average of 9601.62 per m2 3D densified points. An ultra high-resolution (UHD) of 1.05 cm per pixel DSM for a 34,000 m2 coverage was developed. Post-processing of the images was conducted in PIX4D software, the acquirement of the images performed by an autonomous drone, DJI Matrice M100, with a Zenmuse X3 camera.

Aerial mapping was accompanied by Portable X-Ray Fluorescence (PXRF) analysis with a Niton XL3t GOLDD+ which was performed on a grid with 10 x 10 meters cells on the crater's surface, and 155 measurements were conducted for elemental analysis. Probing the Stefanos crater's surface with the PXRF has revealed an average content in parts per million (ppm) of (Mo) 4.5 ± 2.2, (Zr) 135.9 ± 3.5, (Sr) 213.5 ± 3.8, (U)5.1 ± 3.5, (Rb) 27.8 ± 2.1, (Th) 9.0 ± 2.9, (Pb) 136.1 ± 7.1, (Se) 3.8 ± 2.1, (As) 37.5 ± 6.1, (Zn) 12.0 ± 5.2, (Cu) 23.0 ± 11.1, (Ni) 23.8 ±11.5, (Co) 39.7 ± 16.9, (Fe) 6651.3 ± 91.2, (Mn) 68.3 ± 30.8, (Cr) 31.6 ± 6.8, (V) 42.0 ±15.1, (Ti) 1679.6 ± 58.9, (Sc) 35.3 ± 12.6, (Ca) 3696.1 ± 98.2, (K) 5410.7 ± 149.4, (S) 8919.6 ± 351.3, (Ba) 658.5 ± 37.2, (Cs) 31.5 ± 7.5, (Te) 54.1 ± 23.3, (Sb) 24.5 ± 11.6, (Sn) 18.6 ± 10.7, (Cd) 17.6 ± 10.2, (Ag) 14.1 ± 7.4. Geographical information systems (GIS) analysis and implementation of inverse distance weight prediction mapping using the geostatistical analyst module has shown linear features of contrasting geochemical compositions implying the existence of faulting, possibly active and rooted in the upper hydrothermal system underneath the Stefanos crater.

Introduction

- Stefanos crater is part of the Nisyros active volcano, located at the eastern end of the Hellenic Active Volcanic Arc, in Greece (Figure 1 & 2).
- An active volcanic island is rooted on a hydrothermal system, where sea water and magma interact, and gas emission is manifested through fractures and openings on the surface called fumaroles.
- Monitoring of active volcanoes may promote socio-economic stability and maintain a safe touristic activity in the Eastern Mediterranean Sea [1, 2].
- Hydrothermal activity surfaces on the crater a series of major and minor elements and oxides originating from water-rock reactions.
- High-definition geochemical mapping and drone-scanning (Figure 1) on a crater may reveal active structures such as fractures and joints that may allow identification of vulnerable locations and monitoring of the volcano's activity.

Methods

- **Data**
  - ➢ Autonomous Drone DJI Matrice M100 using an Aerial Camera Zenmuse X3 (1.05 centimeters per pixel resolution). Aerial images of the study area were retrieved and composed in PIX4D).
  - ➢ Portable X-ray Fluorescence (PXRF) measurements at 60 seconds per sample (Table 1).
  - ➢ GPS (Geographical Point System) (Integrated in the drone and PXRF)

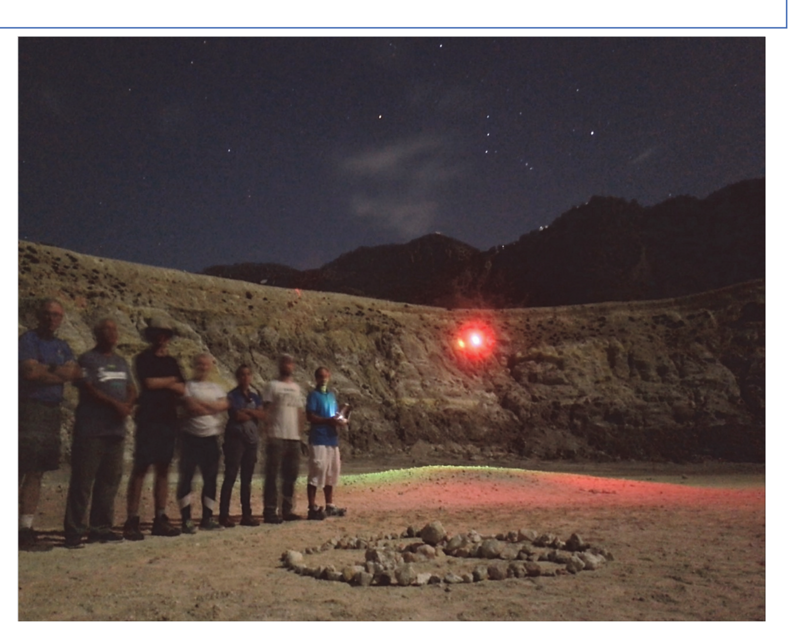


Figure 1: Launching an autonomous drone for automatic thermal night–scanning inside the Stefanos crater on Nisyros Volcano.

- **Techniques & Processing**
  - ➢ Georeferencing of 594 images derived from a day- and night-scanning (Figure 1) at 21 m elevation above the Stefanos crater's surface at Nisyros Volcano
  - ➢ 3D reconstruction and cloud points generation (~ LiDAR) using PIX4D, ArcMap, and Global Mapper to create and analyze DEM (Digital Elevation Model) of the crater
  - ➢ PXRF analysis on 196 sample points (Table 1) on a grid with cells of approximately 10 x 10 meters
  - ➢ Krigering and geochemical mapping using geostatistical analyst extension in ArcMap [3]

Table 1: Probing the Stefanos crater's surface at Nisyros volcano with the PXRF a series of geochemical compositions were determined in parts per million (ppm) for scanning of 60-seconds per sample.

Elements & oxides	Minimum	Maximum parts per million (ppm) (±error)	Mean	Duration *seconds*
Mo	2.74 (±1.77)	8.07 (±3.2)	4.44 (±2.23)	60.00
Zr	40.38 (±2.28)	261.85 (±4.81)	130.56 (±3.47)	60.00
Sr	48.97 (±2.1)	520.36 (±5.81)	219.58 (±3.86)	60.00
U	3.53 (±2.24)	9.94 (±4.96)	5.09 (±3.54)	60.00
Rb	11.14 (±1.62)	68.07 (±2.95)	27.24 (±2.09)	60.00
Th	3.25 (±2.09)	18.78 (±5.13)	9 (±2.98)	60.00
Pb	12.19 (±3.23)	574.02 (±14.04)	146.57 (±7.28)	60.00
Au	0 (±3.63)	0 (±5.34)	<LOD (±4.22)	60.00
Se	3.3 (±1.96)	4.57 (±3.71)	3.83 (±2.87)	60.00
As	9.19 (±3.43)	89.6 (±11.33)	38.42 (±6.25)	60.00
Hg	0 (±8.25)	0 (±13.05)	<LOD (±9.26)	60.00
Zn	6.39 (±4.03)	28.03 (±8.93)	11.84 (±5.31)	60.00
W	0 (±21.6)	0 (±36.21)	<LOD (±24.74)	60.00
Cu	12 (±7.9)	59.88 (±18.91)	23.1 (±11.13)	60.00
Ni	16.52 (±10.65)	40.13 (±20.07)	23.25 (±15.33)	60.00
Co	13.05 (±7.55)	114.35 (±28.41)	39.44 (±16.85)	60.00
Fe	1,058.9 (±39.81)	25,190.77 (±191.17)	6,614.91 (±90.92)	60.00
Mn	39.58 (±25.6)	115.9 (±50.13)	67.24 (±30.79)	60.00
Cr	15.49 (±4.15)	52.72 (±10.31)	31.64 (±6.78)	60.00
V	18.17 (±6.57)	80.87 (±23.76)	42.42 (±15.12)	60.00
Ti	262.52 (±22.37)	2,910.46 (±90.66)	1,673.42 (±58.9)	60.00
Sc	10.84 (±6.05)	144.53 (±30.35)	33.79 (±12.43)	60.00
Ca	437.81 (±45.06)	41,209.41 (±307.54)	3,174.52 (±94.37)	60.00
K	2,075.25 (±76.12)	17,839.84 (±288.51)	5,478.78 (±150.24)	60.00
S	1,754.49 (±147.71)	27,521.7 (±742.62)	9,007.54 (±352.86)	60.00
Ba	182.19 (±29.06)	4,437.77 (±64.13)	663.35 (±37.39)	60.00
Cs	10.95 (±6)	84.49 (±11.41)	32.52 (±7.56)	60.00
Te	28.52 (±17.22)	146.49 (±39.46)	54.35 (±23.11)	60.00
Sb	14.11 (±8.62)	50.35 (±17.57)	24.29 (±11.44)	60.00
Sn	12.13 (±7.51)	36.85 (±19.18)	18.93 (±10.74)	60.00
Cd	13.64 (±8.92)	27.92 (±22.61)	18.36 (±14.15)	60.00
Ag	10.33 (±6.53)	17.86 (±14.98)	13.03 (±9.65)	60.00
Pd	0 (±7.69)	0 (±16.23)	<LOD (±10.06)	60.00
SO3	4,386.21 (±0)	68,804.26 (±0)	22,518.84 (±0)	60.00
K2O	2,511.05 (±0)	21,586.2 (±0)	6,629.33 (±0)	60.00
CaO	612.94 (±0)	57,693.17 (±0)	4,444.32 (±0)	60.00
TiO2	438.41 (±0)	4,860.47 (±0)	2,794.62 (±0)	60.00
Cr2O3	22.62 (±0)	76.97 (±0)	46.19 (±0)	60.00
MnO	14.4 (±0)	149.52 (±0)	77.63 (±0)	60.00
Fe2O3	1,514.23 (±0)	36,022.79 (±0)	9,459.31 (±0)	60.00
NiO	0.05 (±0)	50.97 (±0)	12.15 (±0)	60.00
CuO	0.69 (±0)	74.85 (±0)	19.05 (±0)	60.00
ZnO	1.65 (±0)	35.04 (±0)	11.72 (±0)	60.00

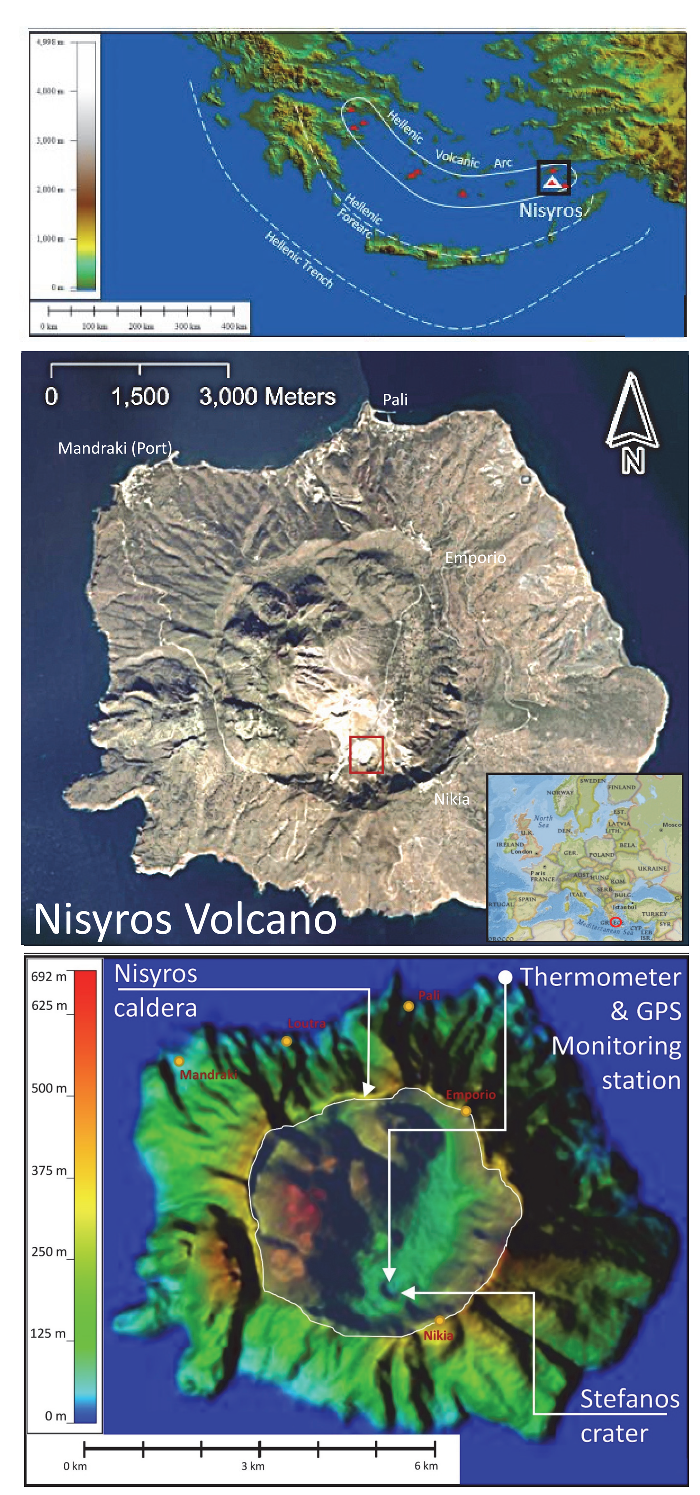
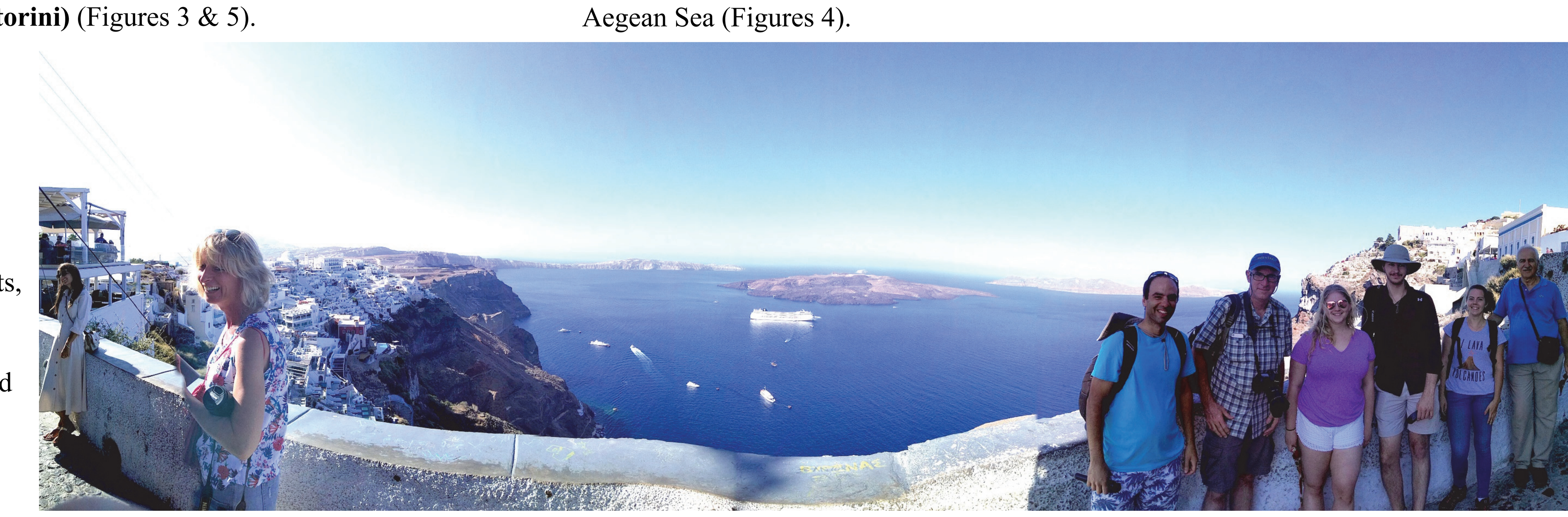


Figure 2: (Upper) South Aegean Sea map showing the formed arcs from the interaction of the subducting African and overriding Aegean micro-plate. (Middle) Satellite image of Nisyros Volcano and (lower) digital surface model (DSM) of the Nisyros volcano, and location of the live monitoring station on Stefanos crater.

Results

- **International field trip and 3D visual representation of the Hellenic Volcanic centers (Nisyros and Santorini)** (Figures 3 & 5).
- **Geochemical GIS analysis of the Stefanos crater at Nisyros Volcano,** South Aegean Sea (Figures 4).

Figure 3: International research team visiting the Greek volcanoes, Santorini and Nisyros islands. (from right to left) Prof. K. Kyriakopoulos from National University of Athens, and Morgan Jones graduate student, Tom Plitnick and Daniella Chernoff undergraduate students, and Prof. Bret Bennington (chair) and Prof. Antonios Marsellos from the department of Geology, Environment and Sustainability at Hofstra University.



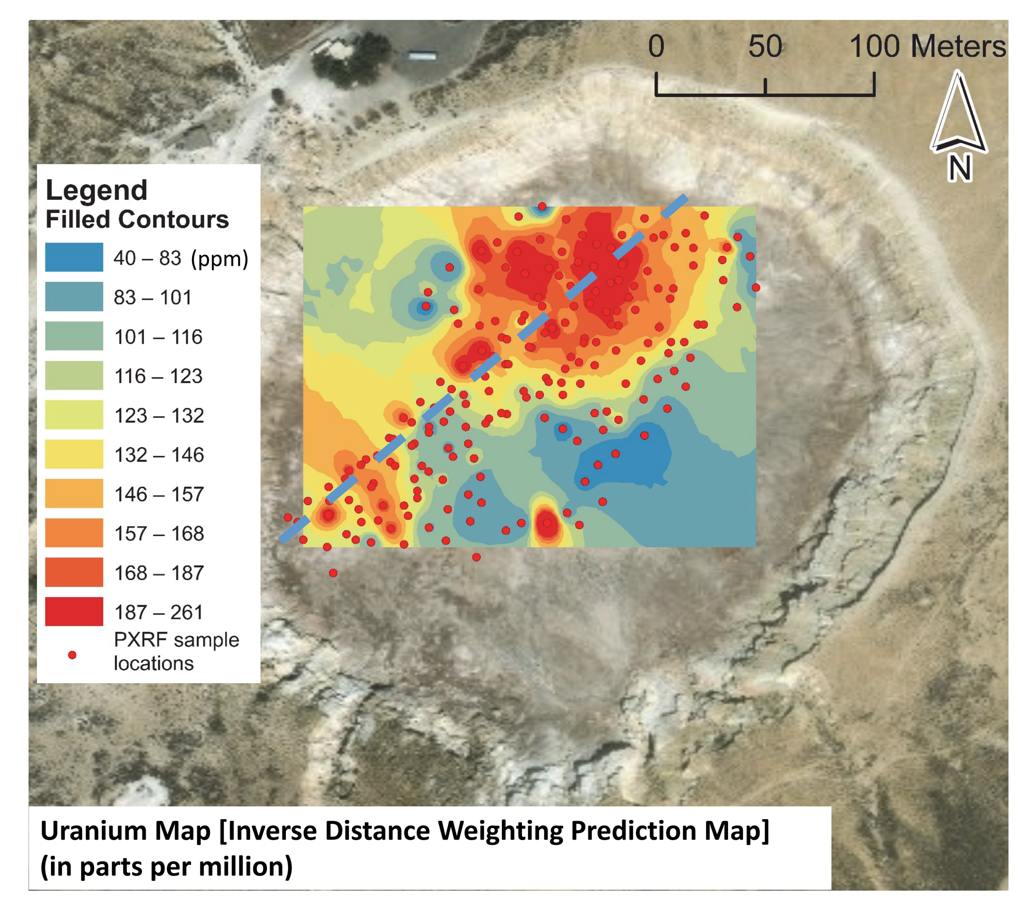


Figure 4: GIS map using inverse distance weighting (implemented with the geostatistical analyst) on uranium content results from the PXRF scanning at the field. Blue dashed line shows a possible linear pattern between the high- and low-concentrations of uranium at the northwestern and southeaster region, respectively.

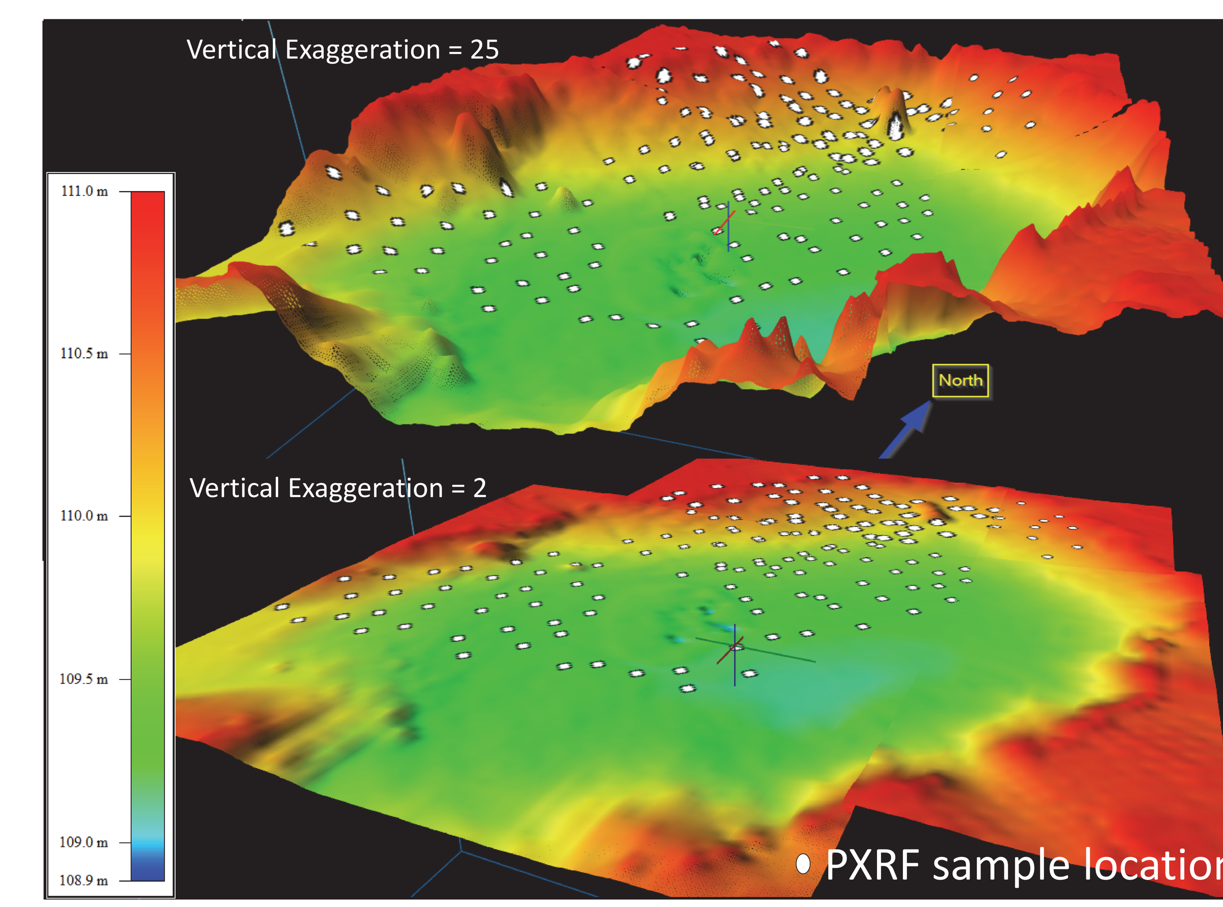


Figure 5: High-definition digital surface model in 3D derived from the autonomous drone scanning at an elevation of 20 m above the crater's surface. White circles are sample locations where PXRF analyses performed at the field. On the volcanic crater, the remaining area with no samples was inaccessible and unsafe with ground temperatures approximately around 100 Celsius (210 Fahrenheit).

Discussion

- **Geochemical data**
  - ➢ Interesting linear patterns were revealed using GIS software on PXRF analysis. NE-trending and NW-trending of contrasting regions between high- and low-concentrations of major and minor elements have been found on the surface of Stefanos crater on Nisyros volcano (Figure 4).
- **Digital Surface Model**
  - ➢ Stefanos crater surface shows a flat geomorphology which may be attributed to surrounding erosion (Figure 5).

References

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Conclusion

- Geographical information systems (GIS) analysis and implementation of inverse distance weight prediction mapping using the geostatistical analyst has shown linear features of contrasting geochemical compositions.
- The existence of faulting, possibly active and rooted in the upper hydrothermal system underneath the Stefanos crater may reflect sharp changes of geochemical compositions on the surface.

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